

RA8P1 E-READER

Processor schematic development
Power networks are being implemented from RA8P1-specific references.
This revision is not a completed schematic or a fabrication release.

RA8P1 IO allocation



File: mcu_interfaces.kicad_sch

Global supplies: +3V3_MCU and GND.
Core regulator output MCU_VCORE stays local to the processor power sheet.

Remaining subsystem circuits follow epic #821 and issues #824-#834:
radio, memory/storage, USB/battery, system rails, display controller,
panel power, touch, warm/cool lighting, buttons and sensors.
No empty subsystem sheets are represented as completed circuits.

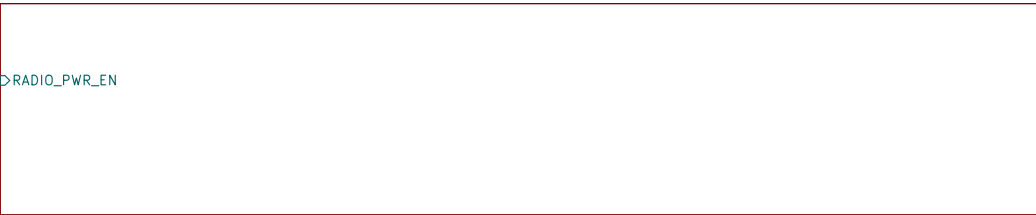
Footprint and PCB qualification are deferred by scope.

RA8P1 internal core regulator



File: mcu_core_power.kicad_sch

ESP32-C6 radio



File: radio_esp32.kicad_sch

RA8P1 IO and analog supplies



File: mcu_supply_power.kicad_sch

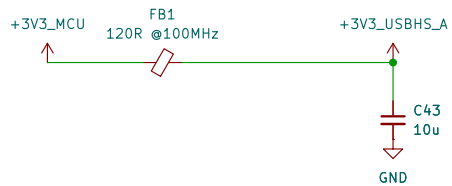
RA8P1 clocks, reset and debug



File: mcu_clocks_debug.kicad_sch

Sheet: / File: ereader_rev1.kicad_sch		
Title: RA8P1 e-reader - schematic index		
Size: A3	Date: 2026-09-05	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.5		Id: 1/6

DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.



USB-004: USB-HS analog supply / FB1 + C43
FB1: 120 ohm at 100 MHz; DCR 0.05 ohm initial max.
Screen with combined USB-HS current: I = 55.3 mA.
Vdrop = I*R = 2.765 mV; P = I^2*R = 0.153 mW.
After-test DCR 0.10 ohm; 5.53 mV / 0.306 mW.
C43: 10 uF +/-10%, 35 V X7R; initial 9.11 uF.
TDK nominal bias curve: 9.644 uF at 3.3 V.
Not transient or all-corners qualification; verify rail ramp.
Full calculations: ../design/usb_interface.md [USB-004]

U1B R7KA8P1KFLCAC#UC0		
GPIO P0 / P1		
U13	P000	P100
T13	P001	P101
U14	P002	P102
R12	P003	P103
T11	P004	P104
R11	P005	P105
U12	P006	P106
T12	P007	P107
U11	P008	P108
P11	P009	P109
P10	P010	P110
N10	P011	P111
P9	P012	P112
N9	P013	P113
T8	P014	P114
U8	P015	P115

U1C R7KA8P1KFLCAC#UC0		
GPIO P2 / P3		
C5	P200	P300
B15	P206	P301
A17	P207	P302
	P303	
	P304	
	P305	
	P306	
	P307	
	P308	
	P309	
	P310	
	P311	
	P312	

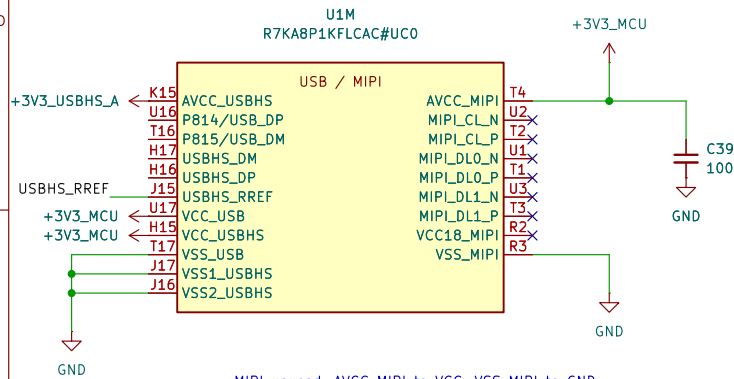
U1D R7KA8P1KFLCAC#UC0		
GPIO P4 / P5		
P17	P400	P500
N17	P401	P501
L14	P402	P502
H13	P403	P503
J13	P404	P504
G12	P405	P505
F14	P406	P506
R17	P407	P507
M15	P408	P508
R16	P409	P509
K13	P410	P510
M14	P411	P511
M12	P412	P512
P14	P413	P513
L13	P414	P514
R15	P415	P515

U1E R7KA8P1KFLCAC#UC0		
GPIO P6 / P7		
P3	P600	P700
P4	P601	P701
P2	P602	P702
P1	P603	P703
N2	P604	P704
N1	P605	P705
N3	P606	P706
M2	P607	P707
K2	P608	P708
A1	P609	P709
D3	P610	P710
D2	P611	P711
E4	P612	P712
C2	P613	P713
E3	P614	P714
E2	P615	P715

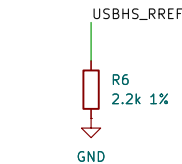
U1F R7KA8P1KFLCAC#UC0		
GPIO P8 / P9		
T6	P800	P902
P6	P801	P903
R6	P802	P904
P7	P803	P905
R7	P804	P906
P12	P805	P907
T14	P806	P908
N11	P807	P909
U5	P808	P910
T5	P809	P911
M9	P810	P912
N8	P811	P913
P8	P812	P914
B1	P813	P915

U1G R7KA8P1KFLCAC#UC0		
GPIO PA / PB		
G1	PA00	PB00
H4	PA01	PB01
F1	PA02	PB02
G2	PA03	PB03
D1	PA04	PB04
G3	PA05	PB05
C1	PA06	PB06
G4	PA07	PB07
F3	PA08	
F2	PA09	
F4	PA10	
B4	PA11	
B2	PA12	
C3	PA13	
D4	PA14	
E1	PA15	

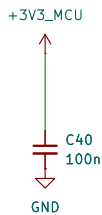
U1H R7KA8P1KFLCAC#UC0		
GPIO PC / PD		
M1	PC00	PD00
M3	PC01	PD01
L2	PC02	PD02
L1	PC03	PD03
L3	PC04	PD04
L5	PC05	PD05
N4	PC06	PD06
K5	PC07	PD07
M4	PC08	
L4	PC09	
M5	PC10	
H5	PC11	
G5	PC12	
J5	PC13	
F5	PC14	
K1	PC15	



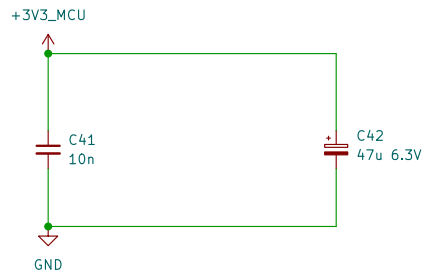
MIPI unused: AVCC_MIPI to VCC; VSS_MIPI to GND.
Leave VCC18_MIPI and all six D-PHY lanes open.
Firmware: MSTPCRC.MSTPC10=1 and MSTPC17=1.
RA8P1 HUM Rev.1.30, section 21.4, p.862.



USB-001: R6 USBHS current-reference resistor
Renesas QDG Table 2, p.7: 2.2 kohm +/-1%.
Rmin = 2200*(1-0.01) = 2178 ohm
Rmax = 2200*(1+0.01) = 2222 ohm
Initial tolerance only; not a derived PHY current.
Full basis + Python check: design/usb_interface.md



USB-002: USB full-speed supply / U1,U17, C40.
Renesas QDG Table 1, p.6: 3.3 V with 100 nF bypass.
C40 = 100*(1 +/-0.10) = 90..110 nF initial; 50 V X7R.
Nominal DC-bias estimate: 99.45 nF at 3.3 V (PWR-001).
Place C40 close to VCC_USB with a short VSS_USB return.
Full basis + Python check: design/usb_interface.md



USB-003: USB-HS supply / U1,H15, C41, C42
Renesas QDG Table 1 p.6: 10 nF ceramic + 47 uF electrolytic.
C41 = 10*(1 +/-0.10) = 9..11 nF initial; 50 V X7R.
TDK nominal bias model: C41(3.3 V) = 10.084 nF (not guaranteed).
C42 = 47*(1 +/-0.20) = 37.6..56.4 uF initial; 6.3 V.
Voltage screening: 3.6/6.3*100 = 57.14% rating, not rail sign-off.
C42 positive to 3V3, negative to GND; no reverse voltage.
C41 close to H15; short returns to VSS1/VSS2_USBHS.
Full math + Python / qualification: ../design/usb_interface.md

PWR-001: C39 AVCC_MIPI bypass, 100 nF (QDG Table 2, p.7).
TDK C1608X7R1H104K080AA; also used at C6 and C16-C34.
Initial C = 100*(1 +/-0.10) = 90..110 nF; 50 V X7R.
TDK DC-bias samples: 99.5575 nF / 3.15 V; 98.9525 nF / 4 V.
C(3.3 V) = 99.5575 + (3.3-3.15)/0.85*(-0.605) = 99.45 nF.
Interpolated nominal reference, NOT a guaranteed minimum.
Final rail / PDN qualification open; 3.3/50*100 = 6.6% rating.
Full basis + Python check: design/power_decoupling.md

RA8P1 e-reader

Sheet: /RA8P1 IO allocation/
File: mcu_interfaces.kicad_sch

Title: RA8P1 IO allocation

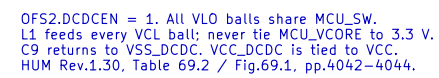
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Date: 2026-09-05

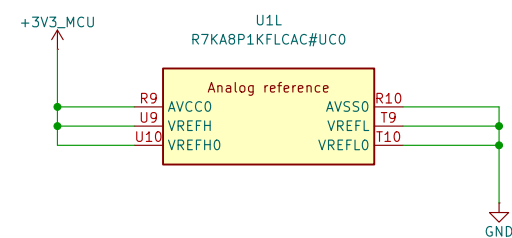
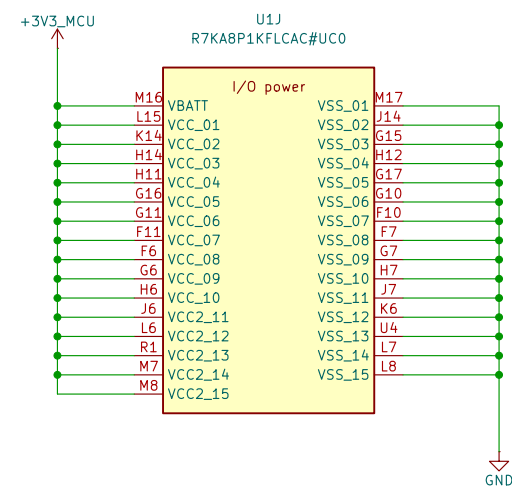
Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.5

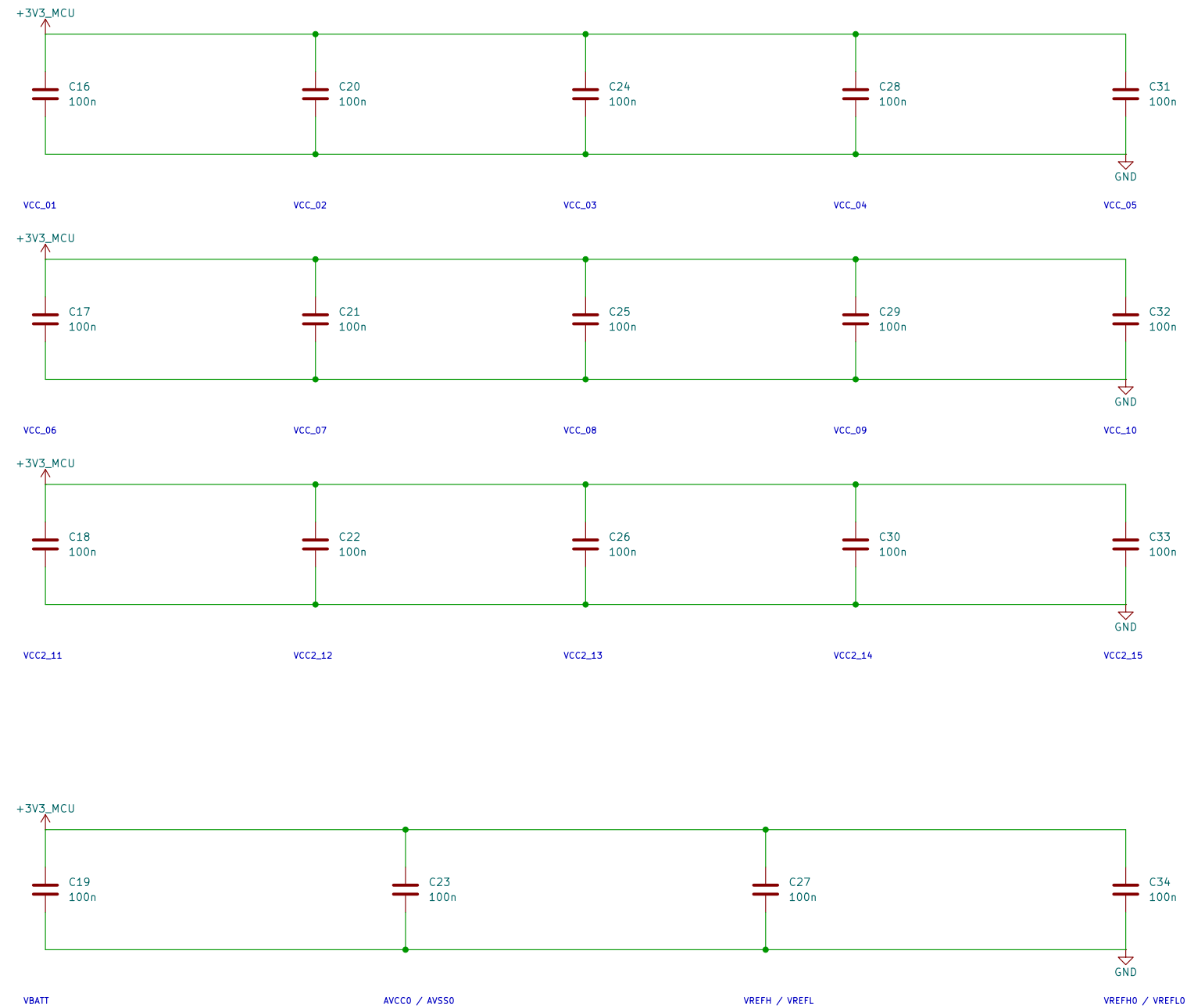
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L1 DCR <= 100mOhm. Effective capacitance and inductor current rating require final MPN qualification. The 3.3 V source and return will be declared on the system-power sheet; no blanket ERC flags.



I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR



VCC and VCC2 use 3.3 V. All returns use one board ground plane. VBATT follows the main rail; no separate backup-cell supply. Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs. USB/MIP1 supplies remain on the interface sheet and are not yet wired. HUM Rev.1.30; Table 69.2 p.4042; unused references: section 21.4 pp.861–862. This is an electrical draft; capacitor MPN qualification remains open.

